

SYLLABUS

BCO 081B	PROGRAMMING WITH PYTHON	3-0-0 [3]
UNIT 1	Introduction: Features of Python, History of Python, installing Python; basic syntax, interactive shell, editing, saving, and running a script. Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types (Numbers, Booleans, Strings, None), Indentation and comments, Type Conversions, type() Function, membership operator	
UNIT 2	Control Flow Statements: if, if-else, if-elif-else, while loop, range() Function, for loop, continue and break statements. Strings: Creating and Storing Strings, Basic String Operations, String Slicing and Joining, String Methods, Formatting Strings	
UNIT 3	Lists, Tuples, Dictionaries, Sets: Basic Operations, Indexing and Slicing, Built-In Functions, Using zip() Function Functions: Function Definition and Calling the Function, The return Statement and void Function, Parameters and Arguments in a Function: Positional, Keyword, Default, Scope and Lifetime of Variables, the return Statement, returning multiple values, recursive functions, Lambda functions	
UNIT 4	Object-Oriented Programming: classes, objects, the self-parameter and adding methods to a class, display class attributes and methods, __init__ method (constructor), __del__ method (destructor), using private instance variables and methods, inheritance, types of inheritance Regular Expression Operations: using r prefix for regular expressions, using parentheses in regular expressions, compiling regular expressions using compile() method of re module, named groups	
UNIT 5	Graphics Programming: The Turtle Module: moving the turtle in any direction, draw different shapes, the color, bgcolor, circle and speed method of turtle, drawing basic shapes using iterations The Tkinter Module: Tk-the GUI Window, displaying images, tkinter Widgets: button, entry, text, canvas, frame, etc.	

Text Book:

Introduction to Python Programming, Gowrishankar S., Veena A., CRC Press, Taylor & Francis Group, ISBN: 978-0-8153-9437-2

Reference Books:

1. Python: Real World Machine Learning, Prateek Joshi et al., Packt Publishing, ISBN 13: 9781787123212
2. Programming and Problem Solving with Python, Ashok Namdev Kamthane et. al., McGraw Hill Education (India) Private Limited, ISBN: 978-93-87067-57-8
3. Fundamentals of Python: First Programs, Kenneth Lambert, Course Technology, Cengage Learning, 2012 ISBN-13: 978-1-111-82270-5